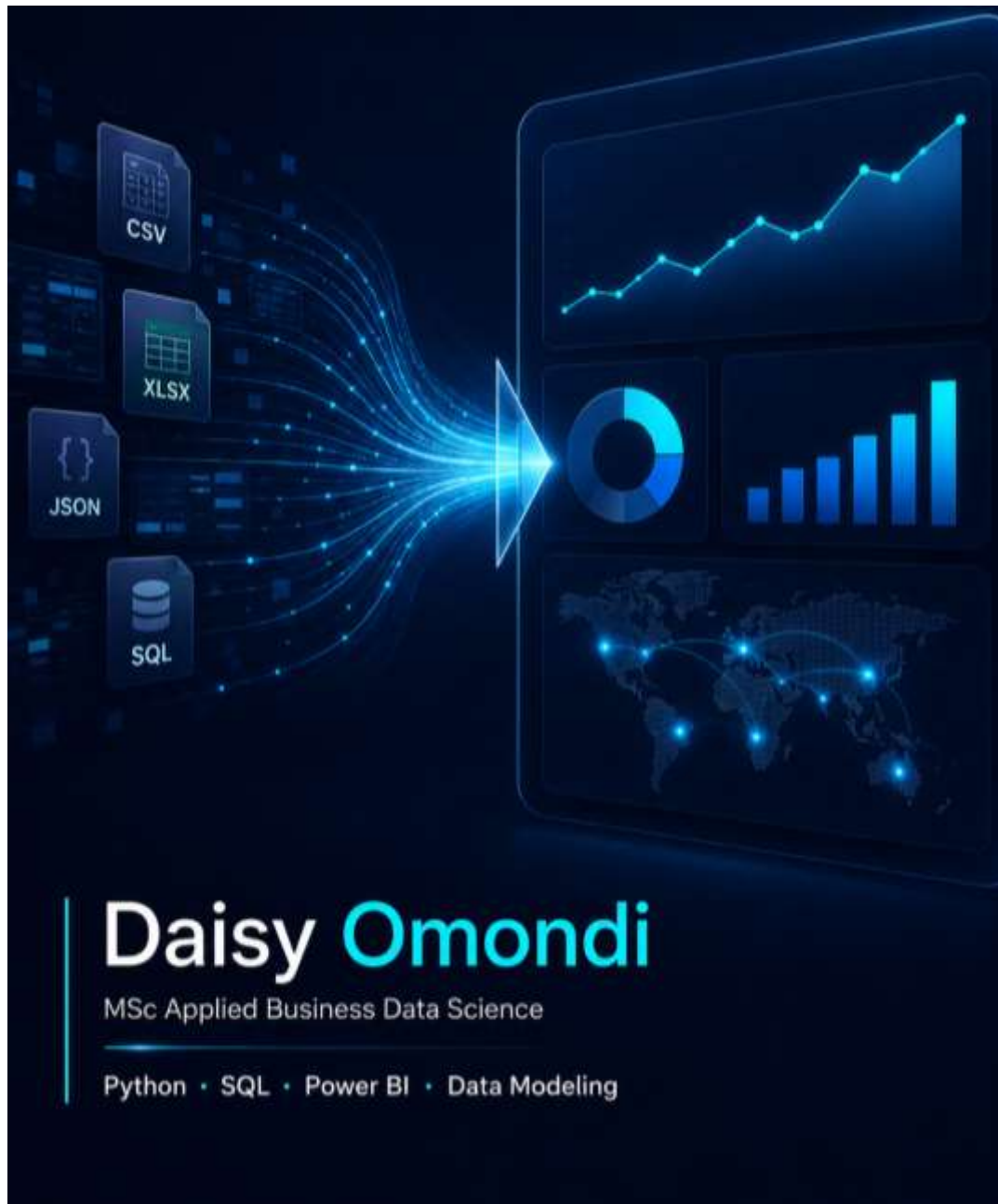


Business KPI Dashboard & Data Modelling

Building a single source of Truth for Business Performance



1. Project Focus

Nova is a growing retail business with sales across different regions, product groups and customer segments. A fictional mid-sized European retail company used for portfolio demonstration purposes. The company had data, but the reporting process was messy. Sales data, customer data and product data were stored in different CSV files. Reports were prepared manually in Excel. This made reporting slow and sometimes confusing.



My goal was to build one clean reporting system. I used Python to clean the raw data. I loaded the cleaned data into SQL. I designed a relational data model. Then I built a Power BI dashboard to help management track the most important KPIs.

I wanted this project to show more than technical skills. I wanted to show how I think. I start with the business problem, clean the data properly, build a strong structure, and then turn the result into insights people can actually use.

Expected business value:

- Better data quality
- Faster reporting
- Clearer KPIs
- Less manual work
- Better decisions for management

2. The Reporting Problem

Nova Retail Group was growing, but the reporting process did not grow with the company. Different teams used different spreadsheets. Some files had missing values. Some dates were not formatted correctly. Some product and customer names were written in different ways. This made the reports harder to trust.

The main problems were:

- Reporting took too much time
- KPI definitions were not always clear
- Manual Excel work created room for mistakes
- Managers could not quickly compare regions, products and customer groups
- The process was not ready for future growth

The company did not need more spreadsheets, rather it needed a better reporting foundation.



My project goals were:

- Clean and prepare raw business data with Python
- Build a structured SQL database
- Create a relational data model
- Define clear KPIs
- Build an interactive Power BI dashboard
- Turn the results into business insights

3. BI Data Pipeline Architecture

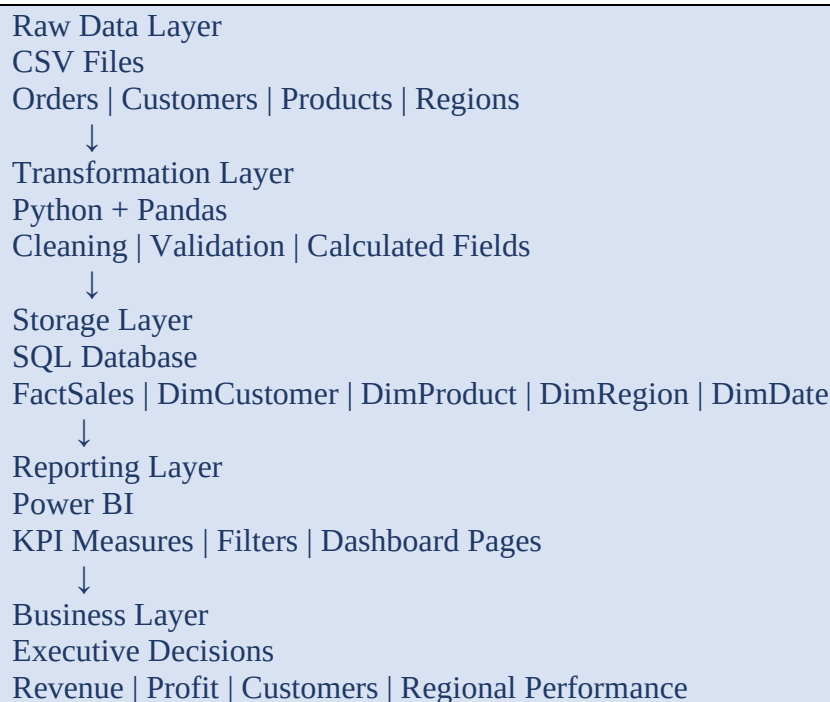
I started with raw CSV files that represented typical business data:

- Orders
- Customers
- Products
- Regions
- Dates
- Sales transactions

Before building the dashboard, I cleaned the data with Python and Pandas. This step mattered because a dashboard is only as good as the data behind it. The cleaning work included:

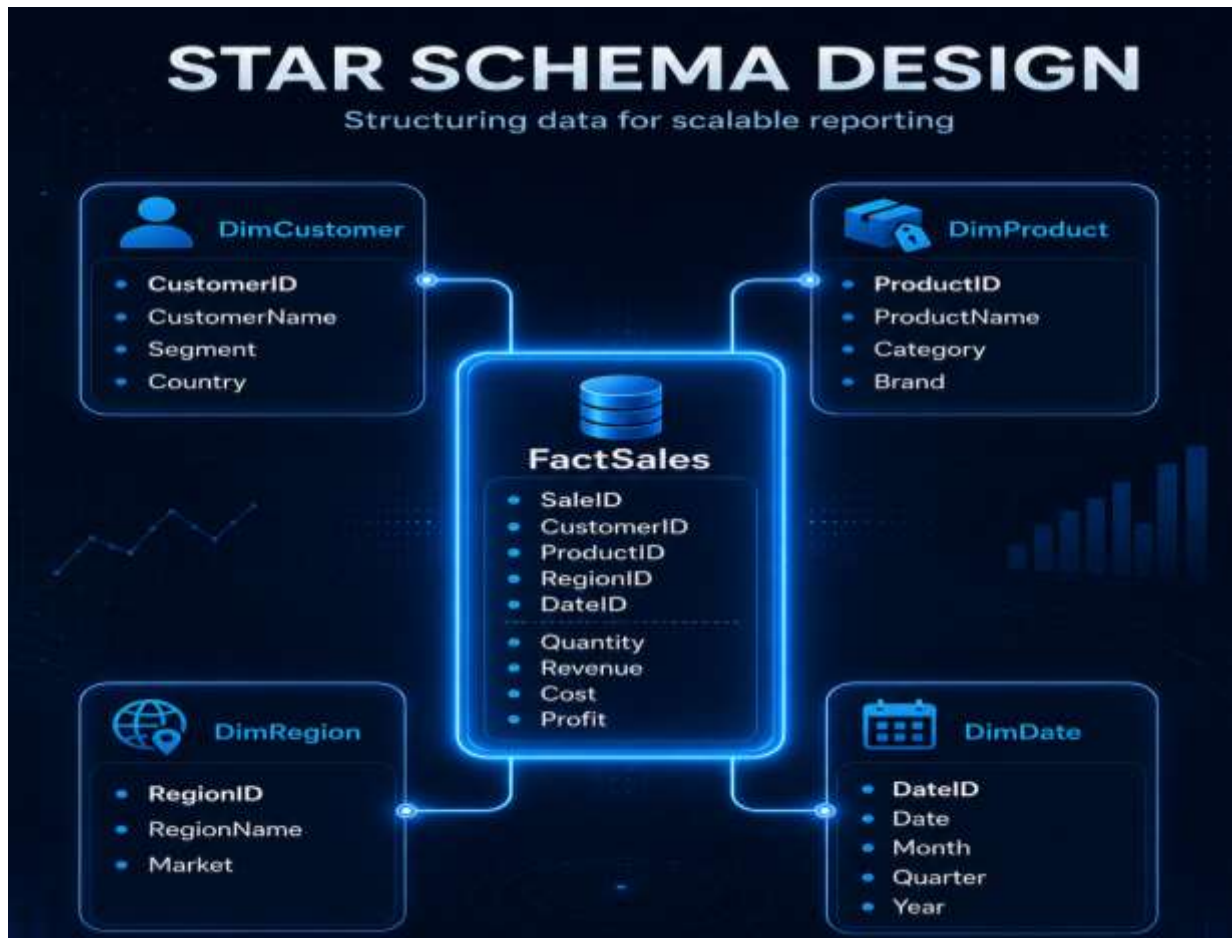
- Removing duplicate records
- Checking missing values
- Standardizing date formats
- Renaming columns clearly
- Cleaning inconsistent text values
- Creating calculated fields like revenue, cost, profit and profit margin
- Checking that customer, product and order IDs matched correctly

After cleaning the data, I created an ETL workflow. The raw files were extracted from CSV. The data was transformed in Python. The cleaned data was loaded into SQL. Power BI then connected to the structured database. This made the process cleaner and easier to repeat. My healthcare operations background helped me here. In healthcare, details matter. A small mistake can create a big problem. I bring that same mindset into data work. I do not rush to the dashboard. I first make sure the foundation is clean.



4. Data Model Design

After cleaning the data, I designed a relational data model. I did not keep everything in one large table. That would have made the reporting messy and harder to maintain. Instead, I used a star schema with one fact table and several dimension tables. The main table was FactSales.



Each dimension table had a primary key. The FactSales table used foreign keys to connect the tables. This model made the dashboard stronger. It allowed management to answer questions like:

- Which region brings the most revenue?
- Which product category has the best profit margin?
- How does revenue change over time?
- Which customer segment creates the highest value?

5. Power BI Dashboard

The final output was an interactive Power BI dashboard for business reporting. The dashboard was designed for management. It gives a clear overview of business performance without forcing the user to search through many files.

The top section shows the main KPIs:

Total Revenue
Total Profit
Profit Margin
Number of Orders
Average Order Value
Customer Count

The dashboard also includes charts for sales trends, product performance, regional performance and customer segments. The revenue trend helps management see how sales change over time. The product view shows which products and categories perform best. The regional view shows where the business is strong and where it needs more attention. The customer view helps management understand which customer groups create more value.

I also added filters for year, region, product category and customer segment. This allows the user to explore the data based on their own questions. I kept the dashboard simple on purpose. My goal was not to impress people with too many charts. My goal was to make the numbers easy to understand.



6. Business Insights

The dashboard helped turn raw data into useful business insights. The first insight was that revenue was not equal across all regions. Some regions performed strongly, while others had weaker sales and lower profit. This means the company should not use the same sales strategy everywhere.

The second insight was that some product categories had high revenue but lower profit margins. This is important because high sales do not always mean strong profit.

The third insight was that a small group of products created a large part of total revenue. These products should be watched closely because they have a strong effect on business performance.

The fourth insight was that customer segments behaved differently. Some customers bought more often. Others spent more per order. This can help the company create better marketing and retention actions.

My recommendations:

- Review underperforming regions
- Check prices and costs for low margin products
- Monitor top products more closely
- Use customer segments for targeted campaigns
- Replace manual monthly reporting with the dashboard process
- Keep improving the data model as the company grows

Business impact: The company gets faster reporting, clearer KPIs and better decision support.

 Insight	 Meaning	 Action
 Regional sales vary	Some regions perform much better than others.	 Review weak regions and adjust the sales strategy.
 High sales, lower margin	Some categories sell well but bring less profit.	 Check pricing, discounts and product costs.
 A few products drive revenue	A small product group has a strong impact on total sales.	 Protect top products and monitor them closely.
 Customer groups behave differently	Segments differ in spending and buying patterns.	 Use targeted campaigns for each segment.

7. Skills & Technical Summary

This project shows the full business intelligence workflow from raw data to business insight. I used Python for data cleaning and transformation. I used SQL to structure the cleaned data and create a proper reporting layer. I used Power BI to build the dashboard and communicate the KPIs clearly.

